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|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Power (2 engines):</b><br>W_engine = 220 kg<br>P_sl = 2100 hp = 1540 kW<br>nu = 0.8<br>P_available = nu * P_hp = 1232 kW | <b>Source:</b><br><a href="http://www.iascanada.com/engmodels.html#:~:text=Table_content:%20header:%20%7C%20Model%20Number%20%7C%20Shaft,Shaft%20H/P:%20850%20%7C%20Weight(lbs):%20429%20%7C">http://www.iascanada.com/engmodels.html#:~:text=Table_content:%20header:%20%7C%20Model%20Number%20%7C%20Shaft,Shaft%20H/P:%20850%20%7C%20Weight(lbs):%20429%20%7C</a><br><a href="http://www.iascanada.com/engmodels.html#:~:text=Table_content:%20header:%20%7C%20Model%20Number%20%7C%20Shaft,Shaft%20H/P:%20850%20%7C%20Weight(lbs):%20429%20%7C">http://www.iascanada.com/engmodels.html#:~:text=Table_content:%20header:%20%7C%20Model%20Number%20%7C%20Shaft,Shaft%20H/P:%20850%20%7C%20Weight(lbs):%20429%20%7C</a><br>Conservative estimate<br>Derived |
| <b>Wing:</b><br>S = 520 ft^2 = 48 m^2<br>AR = 9.5<br>b = sqrt(AR*S) = 70.3 ft = 21.4 m                                      | Chatgpt Guess<br>Chatgpt Guess<br>Derived                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Aero:</b><br>Cd0 = 0.028<br>Cl_max = 1.9                                                                                 | Chatgpt Guess<br>My Guess based on past reports and hw examples                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Other</b><br>e = 0.9<br>W_tot = 14500 lbs = 6580 kg                                                                      | Given                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |